

Two Phase Method

The **Big-M method** uses a very large penalty M to eliminate artificial variables from the final solution. But in practice, choosing and handling such a “very large number” may cause **computational difficulties**:

- Arithmetic overflow or instability for large M .
- Ambiguity when comparing terms involving M .
- Unnecessary complexity in bookkeeping.

To avoid these drawbacks, the **Two-Phase Method** was developed.

- **Phase I:** Ignore the original objective function. Instead, minimize the sum of artificial variables.
 - If the minimum value = 0, then a feasible solution to the original problem exists.
 - If > 0 , the original LP is **infeasible**.
- **Phase II:** Use the feasible solution from Phase I as the starting basic feasible solution (BFS), then solve the original LP with the usual simplex method.

Solve the following LPP by two-phase method:

$$\text{Max } Z = -4x_1 - x_2$$

$$\text{Subject to: } 3x_1 + x_2 = 3,$$

$$4x_1 + 3x_2 \geq 6,$$

$$x_1 + 2x_2 \leq 4, x_1, x_2 \geq 0.$$

Solution: : Let us introduce the **surplus variable** s_1 in the second constraint, **slack variable** s_2 in the third constraint and **artificial variables** A_1, A_2 in the first and second constraints, respectively.

$$\text{Max } Z = -4x_1 - x_2 + 0.s_1 + 0.s_2 - A_1 - A_2$$

$$\text{Subject to: } 3x_1 + x_2 + A_1 = 3,$$

$$4x_1 + 3x_2 - s_1 + A_2 = 6,$$

$$x_1 + 2x_2 + S_2 = 4, \quad x_1, x_2, s_i, A_i \geq 0.$$

For Phase I we consider **only the artificial variables in the objective function:**

$$\text{Max } Z = -A_1 - A_2$$

$$\text{Subject to: } 3x_1 + x_2 + A_1 = 3,$$

$$4x_1 + 3x_2 - s_1 + A_2 = 6,$$

$$x_1 + 2x_2 + S_2 = 4, \quad x_1, x_2, s_i, A_i \geq 0.$$

First Phase table:

		$C_j \rightarrow$	0	0	0	0	-1	-1	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	A_1	A_2	
-1	A_1	3	3	1	0	0	1	0	1 $\rightarrow$
-1	A_2	6	4	3	-1	0	0	1	3/2
0	s_2	4	1	2	0	1	0	0	4
$Z_j - C_j \rightarrow$		$Z = 0$	-7 $\uparrow$	-4	1	0	0	0	
0	x_1	1	1	1/3	0	0	1/3	0	3
-1	A_2	2	0	5/3	-1	0	-4/3	1	6/5 $\rightarrow$
0	s_2	3	0	5/3	0	1	-1/3	0	9/5
$Z_j - C_j \rightarrow$		$Z = 9$	0	-5/3 $\uparrow$	1	0	7/3	0	
0	x_1	3/5	1	0	1/5	0	3/5	-4/5	
0	x_2	6/5	0	1	-3/5	0	-4/5	3/5	
0	s_2	1	0	0	10	1	1	-1	
$Z_j - C_j \rightarrow$		$Z = 11$	0	0	0	0	1	1	

Here $Z_j - C_j \geq 0, \forall j$ and the value of the objective function is zero, so the LPP has feasible solution and we proceed to Phase II for the problem.

Second Phase table:

		$C_j \rightarrow$	-4	- 1	0	0	Min Ratio $\frac{b}{a_{ij}(> 0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	
-4	x_1	3/5	1	0	1/5	0	3
-1	x_2	6/5	0	1	- 3/5	0	-
0	s_2	1	0	1	1	1	1 $\rightarrow$
$Z_j - C_j \rightarrow$		$Z = 0$	0	0	- 1/5 $\uparrow$	0	
-4	x_1	2/5	1	0	0	- 1/5	1 $\rightarrow$
-1	x_2	9/5	0	1	0	3/5	5
0	s_1	1	0	1	1	1	
$Z_j - C_j \rightarrow$		$Z = -17/5$	0	0	0	1/5	

Here $Z_j - C_j \geq 0, \forall j$, so the optimality condition reached.

The optimal solution is:

$x_1 = \frac{3}{2}, \quad x_2 = \frac{9}{5}, \quad Z_{max} = -\frac{17}{5}$

Ex: Solve the following LPP by two-phase method:

$$\text{Max } Z = 2x_1 + 2x_2 + 4x_3$$

$$\text{Subject to: } 2x_1 + x_2 + x_3 \leq 2,$$

$$3x_1 + 4x_2 + 2x_3 \geq 8, \quad x_1, x_2 \geq 0.$$

Solution: The standard of the given LPP:

$$\text{Max } Z = 2x_1 + 2x_2 + 4x_3 + 0.s_1 + 0.s_2 - A_1$$

$$\text{Subject to: } 2x_1 + x_2 + x_3 + s_1 = 2,$$

$$3x_1 + 4x_2 + 2x_3 - s_2 + A_1 = 8, \quad x_1, x_2, s_i, A_1 \geq 0.$$

For Phase I we consider **only the artificial variables in the objective function:**

$$\text{Max } Z = -A_1$$

$$\text{Subject to: } 2x_1 + x_2 + x_3 + s_1 = 2,$$

$$3x_1 + 4x_2 + 2x_3 - s_2 + A_1 = 8, \quad x_1, x_2, s_i, A_1 \geq 0.$$

Table for phase-I:

		$C_j \rightarrow$	0	0	0	0	0	-1	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	x_3	s_1	s_2	A_1	
0	s_1	2	2	1	1	1	0	0	2 $\rightarrow$
-1	A_1	8	3	4	2	0	-1	1	2
$Z_j - C_j \rightarrow$			-3	-4 $\uparrow$	-2	0	1	0	
0	x_2	2	2	1	1	1	0	0	
-1	A_1	0	-5	0	-2	-4	-1	1	$\rightarrow$
$Z_j - C_j \rightarrow$			5 $\uparrow$	0	2	4	1	0	
0	x_2	2	0	1	1/5	-3/5	-2/5	2/5	
0	x_1	0	1	0	2/5	4/5	1/5	-1/5	
$Z_j - C_j \rightarrow$			0	0	0	0	0	1	

Whenever there is a tie in the leaving variables, the next iteration always give a degenerate solution.

If there is a tie in the min ratio, choose any one

Pivot Row

As the value of the artificial variable is zero, replace that by a non basic non artificial variable, whose coefficient is nonzero in the pivot row.

		$C_j \rightarrow$	2	2	4	0	0	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	x_3	s_1	s_2	
2	x_2	2	0	1	$1/5$	$-3/5$	$-2/5$	10
2	x_1	0	1	0	$2/5$	$4/5$	$1/5$	$0 \rightarrow$
$Z_j - C_j \rightarrow$		$Z = 4$	0	0	$-14/5 \uparrow$	$2/5$	$-2/5$	
2	x_2	2	$-1/2$	1	0	-1	$-1/2$	$1 \rightarrow$
2	x_3	0	$5/2$	0	1	2	$1/2$	5
$Z_j - C_j \rightarrow$		$Z = 4$	7	0	0	6	1	

Here $Z_j - C_j \geq 0, \forall j$, so the optimality condition reached. The optimal solution is:
 $x_1 = 0, \quad x_2 = 2, \quad x_3 = 0, \quad Z_{max} = 4$ **(Degenerate Basic Feasible Solution)**

Special Case: When the artificial variables are at zero level but there exist no non-basic variables(non artificial) with nonzero coefficient in the pivot row.

Ex: Solve the following LPP by two-phase method:

$$\text{Max } Z = 3x_1 + 2x_2 + 3x_3$$

$$\text{Subject to: } 2x_1 + x_2 + x_3 = 2,$$

$$x_1 + 3x_2 + x_3 = 6,$$

$$3x_1 + 4x_2 + 2x_3 = 8, \quad x_1, x_2, x_3 \geq 0.$$

Solution: The standard of the given LPP:

$$\text{Max } Z = 3x_1 + 2x_2 + 3x_3 - A_1 - A_2 - A_3$$

$$\text{Subject to: } 2x_1 + x_2 + x_3 + A_1 = 2,$$

$$x_1 + 3x_2 + x_3 + A_2 = 6,$$

$$3x_1 + 4x_2 + 2x_3 + A_3 = 8, \quad x_1, x_2, x_3, A_i \geq 0.$$

For Phase I we consider only the artificial variables in the objective function:

$$\text{Max } Z = -A_1 - A_2 - A_3$$

$$\text{Subject to: } 2x_1 + x_2 + x_3 + A_1 = 2,$$

$$x_1 + 3x_2 + x_3 + A_2 = 6,$$

$$3x_1 + 4x_2 + 2x_3 + A_3 = 8, \quad x_1, x_2, x_3, A_i \geq 0.$$

		$C_j \rightarrow$	0	0	0	-1	-1	-1	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	x_3	A_1	A_2	A_3	
-1	A_1	2	2	1	1	1	0	0	2 $\rightarrow$
-1	A_2	6	1	3	1	0	1	0	2
-1	A_3	8	3	4	2	0	0	1	2
$Z_j - C_j \rightarrow$			-6	-8 $\uparrow$	-4	0	0	0	
0	x_2	2	2	1	1	1	0	0	
-1	A_2	0	-5	0	-2	-3	1	0	$\rightarrow$
-1	A_3	0	-5	0	-2	-4	0	1	
$Z_j - C_j \rightarrow$			10 $\uparrow$	0	4	8	0	0	
0	x_2	2	0	1	1/5	-3/5	-2/5	2/5	
0	x_1	0	1	0	2/5	3/5	-1/5	0	
-1	A_3	0	0	0	0	-1	-1	1	
$Z_j - C_j \rightarrow$			0	0	0	2	2	1	

In the final table of Phase I, the artificial variable A_3 present as a basic variable with value zero and none of the non-basic (non-artificial) variable exist with nonzero coefficient in the pivot row. Therefore, we remove the corresponding row and columns of the artificial variable in the initial table of Phase II.

Table for Phase II:

		$C_j \rightarrow$	3	2	3	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	x_3	
2	x_2	2	0	1	1/5	10
3	x_1	0	1	0	2/5	0 $\rightarrow$
$Z_j - C_j \rightarrow$		$Z = 0$	0	0	-7/5 $\uparrow$	
2	x_2	2	-1/2	1	0	
3	x_3	0	5/2	0	1	
$Z_j - C_j \rightarrow$		$Z = 4$	7/2	0	0	

Here $Z_j - C_j \geq 0, \forall j$, so the optimality condition reached. The optimal solution is:
 $x_1 = 0, \quad x_2 = 2, \quad x_3 = 0, \quad Z_{max} = 4$ **(Degenerate Basic Feasible Solution)**

Alternate solution case:

Ex: Solve the following LPP by simplex method:

$$\text{Max } Z = 2x_1 + 4x_2$$

$$\text{Subject to: } x_1 + 2x_2 \leq 5,$$

$$x_1 + x_2 \leq 4, \quad x_1, x_2 \geq 0$$

Solution: The standard form of the given LPP:

$$\text{Max } Z = 2x_1 + 4x_2 + 0.s_1 + 0.s_2$$

$$\text{Subject to: } x_1 + 2x_2 + s_1 = 5,$$

$$x_1 + x_2 + s_2 = 4, \quad x_1, x_2, s_1, s_2 \geq 0$$

		$C_j \rightarrow$	2	4	0	0	Min Ratio $\frac{b}{a_{ij}(>0)}$
C_B	B. V.	$b_i \downarrow$	x_1	x_2	s_1	s_2	
0	s_1	5	1	2	1	0	$5/2 \rightarrow$
0	s_2	4	1	1	0	1	4
$Z_j - C_j \rightarrow$		$Z = 0$	-2	-4 $\uparrow$	0	0	
4	x_2	5/2	1/2	1	1/2	0	5
0	s_2	3/2	1/2	0	-1/2	1	$3 \rightarrow$
$Z_j - C_j \rightarrow$		$Z = 10$	0 $\uparrow$	0	2	0	
4	x_2	1	0	1	1	-1	
2	x_1	3	1	0	-1	2	
$Z_j - C_j \rightarrow$		$Z = 10$	0	0	2	0	

In table two, $Z_j - C_j \geq 0, \forall j$ so the optimality condition reached and we see that $Z_j - C_j = 0$ for the variable x_1 , which is a non-basic variable. Therefore, the given LPP have multiple optimal solutions. The other solution we can get from table three. Hence, the optimal solutions are:

$$\begin{aligned}
 x_1 = 0, \quad x_2 = \frac{5}{2}, \quad Z_{max} = 10 \\
 x_1 = 3, \quad x_2 = 1, \quad Z_{max} = 10
 \end{aligned}$$

Exercises

Solve the following LPP:

1. $\text{Max } Z = 3x_1 + 9x_2$

Sub. To: $x_1 + 4x_2 \leq 8,$

$x_1 + 2x_2 \leq 4, \quad x_1, x_2 \geq 0.$

2. $\text{Max } Z = 3x_1 + 2x_2$

Sub. To: $4x_1 - x_2 \leq 8,$

$4x_1 + 3x_2 \leq 12,$

$4x_1 + x_2 \leq 8, \quad x_1, x_2 \geq 0.$

3. $\text{Max } Z = 2x_1 + 4x_2 + 4x_3 - 3x_4$

Sub. To: $x_1 + x_2 + x_3 = 4,$

$x_1 + 4x_2 + x_4 = 8, \quad x_1, x_2, x_3, x_4 \geq 0.$

Ex: Consider the following LPP:

$$\text{Max } Z = x_1 - 3x_2 + x_3$$

$$\text{Sub. To: } 3x_1 - x_2 + 3x_3 \leq 7,$$

$$-2x_1 + 4x_2 \leq 12,$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10, \quad x_1, x_2, x_3 \geq 0.$$

Without performing the simplex table find the missing entries of the following simplex table:

B. V.	Solution ↓	x_1	x_2	x_3	A_1	A_2	A_3
x_1	*	1	0	6/5	2/5	1/10	0
x_2	*	*	1	—	1/5	*	0
s_3	11	0	0	—	11	*	−1/2
Z		1	1	*	−1/5	*	0

Solution: By introducing slack variables s_1, s_2, s_3 in the constraints, we get

$$3x_1 - x_2 + 3x_3 + s_1 = 7,$$

$$-2x_1 + 4x_2 + s_2 = 12,$$

$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10$$

The coefficient matrix is given by:

$$A = \begin{bmatrix} 3 & -1 & 3 & 1 & 0 & 0 \\ -2 & 4 & 0 & 0 & 1 & 0 \\ -4 & 3 & 8 & 0 & 0 & 1 \end{bmatrix} = [A_1 \ A_2 \ A_3 \ A_4 \ A_5 \ A_6] \quad \text{and} \quad b = \begin{bmatrix} 7 \\ 12 \\ 10 \end{bmatrix}.$$

From the given simplex table, we can get the set of basic variables is $\{x_1, x_2, s_3\}$.

Therefore, the corresponding B-basis matrix from the given constraints: $B = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & 0 \\ -4 & 3 & 1 \end{bmatrix}$.

$$\text{Then } B^{-1} = \begin{bmatrix} 2/5 & 1/10 & 0 \\ 1/5 & 3/10 & 0 \\ 1 & -1/2 & 1 \end{bmatrix}.$$

Therefore, the corresponding columns of the simplex table is given by the following formula:

$$\text{Column of } x_1 = B^{-1}A_1 = \begin{bmatrix} 2/5 & 1/10 & 0 \\ 1/5 & 3/10 & 0 \\ 1 & -1/2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$

$$\text{Column of } x_3 = B^{-1}A_3 = \begin{bmatrix} 2/5 & 1/10 & 0 \\ 1/5 & 3/10 & 0 \\ 1 & -1/2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 6/5 \\ 3/5 \\ 11 \end{bmatrix}, \text{ Col. of } s_1 = B^{-1}A_4 = \begin{bmatrix} 2/5 \\ 1/5 \\ 1 \end{bmatrix}, \text{ Col. of } s_2 = \begin{bmatrix} 1/10 \\ 3/10 \\ -1/2 \end{bmatrix}.$$

The solution of the basic variables $B^{-1}b = \begin{bmatrix} 2/5 & 1/10 & 0 \\ 1/5 & 3/10 & 0 \\ 1 & -1/2 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 12 \\ 10 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 11 \end{bmatrix}$.

The value of $Z_3 = CB^{-1}A_3 = [1 \ -3 \ 1] \begin{bmatrix} 6/5 \\ 3/5 \\ 11 \end{bmatrix} = -3/5$, Similarly, $Z_5 = CB^{-1}A_5 = -4/5$.

Now we can complete the given table:

B. V.	Solution ↓	x_1	x_2	x_3	A_1	A_2	A_3
x_1	4	1	0	6/5	2/5	1/10	0
x_2	6	0	1	3/5	1/5	3/10	0
s_3	11	0	0	11	1	-1/2	1
Z		1	1	-3/5	-1/5	-4/5	0

Exercises

Consider the following LPP:

$$\text{Max } Z = 4x_1 + x_2 + 3x_3 + 5x_4$$

$$\text{Sub. To: } 4x_1 - 6x_2 - 5x_3 + 4x_4 \geq -20,$$

$$3x_1 - 2x_2 + 4x_3 + x_4 \leq 11,$$

$$8x_1 - 3x_2 + 3x_3 + 2x_4 \leq 23, \quad x_1, x_2, x_3, x_4 \geq 0.$$

Without performing the simplex table find the missing entries of the following simplex table:

B. V.	Solution ↓	x_1	x_2	x_3	x_4	s_1	s_2	s_3
s_1	*	*	0	*	*	1	0	2
x_4	*	*	0	*	*	0	-3	2
x_2	*	*	1	*	*	0	-2	*
Z		33	1	-35	0	0	-17	*